Build: Custom Dropdown / Select

Problem: Build a custom dropdown with keyboard navigation, outside click to close, and accessible markup.

What This Tests: Refs, keyboard events, click-outside pattern, controlled component, accessibility.

jsx

JSX Copy

```jsx
function Dropdown({ options, value, onChange, placeholder = 'Select ... ' })
  const [isOpen, setIsOpen] = useState(false);
  const [highlightedIndex, setHighlightedIndex] = useState(-1);
  const containerRef = useRef(null);

  useEffect(() => {
    const handleClickOutside = (e) => {
      if (containerRef.current && !containerRef.current.contains(e.target)
        setIsOpen(false);
      }
    };
    document.addEventListener('mousedown', handleClickOutside);
    return () => document.removeEventListener('mousedown', handleClickOuts
  }, []);

  const handleKeyDown = (e) => {
    if (!isOpen && (e.key === 'Enter' || e.key === 'ArrowDown')) {
      setIsOpen(true);
      return;
    }
    if (e.key === 'Escape') { setIsOpen(false); return; }
    if (e.key === 'ArrowDown') { setHighlightedIndex(prev => Math.min(prev
```

```
      setIsOpen(false);
    }
  };

  const selectedLabel = options.find(o => o.value === value)?.label || pla

  return (
    <div ref={containerRef} onKeyDown={handleKeyDown} tabIndex={0} style={
      <div onClick={() => setIsOpen(!isOpen)} style={{
        padding: '8px 12px', border: '1px solid #ccc', borderRadius: 4, cu
        display: 'flex', justifyContent: 'space-between',
      }}>
        <span>{selectedLabel}</span>
        <span>{isOpen ? '▲' : '▼'}</span>
      </div>

      {isOpen && (
        <ul style={{
          position: 'absolute', top: '100%', left: 0, right: 0,
          border: '1px solid #ccc', borderRadius: 4, background: 'white',
          listStyle: 'none', margin: 0, padding: 0, maxHeight: 200, overfl
          zIndex: 10,
        }}>
          {options.map((option, index) => (
            <li
              key={option.value}
              onClick={() => { onChange(option); setIsOpen(false); }}
              onMouseEnter={() => setHighlightedIndex(index)}
              style={{
                padding: '8px 12px', cursor: 'pointer',
                backgroundColor: index === highlightedIndex ? '#f0f0f0' :
              }}
            >
              {option.label}
            </li>
          ))}
        </ul>
      )}
    </div>
  );
}
```

```
const options = [
  { value: 'react', label: 'React' },
  { value: 'vue', label: 'Vue' },
  { value: 'angular', label: 'Angular' },
];

function App() {
  const [selected, setSelected] = useState('');
  return <Dropdown options={options} value={selected} onChange={(opt) => s
}
```

Key Concepts: Click-outside with ref and document listener, keyboard navigation (Arrow Up/Down, Enter, Escape), highlighted index tracking, controlled value, tabIndex for focus management.

QUICK CHECK — no marks, just for you.

1. In this Dropdown, how is a click outside the component detected so the menu can close?

- ◯ React's synthetic event system bubbles the click to the root, where onClick compares event.currentTarget
- ◯ A document-level mousedown listener checks whether the ref'd container element contains the event target
- ◯ An onBlur handler on the container div closes the menu whenever focus leaves it
- ◯ A transparent full-screen overlay div is rendered behind the menu and its onClick closes it

Check answer

2. Why does the container div carry tabIndex={0} in this implementation?

So the dropdown menu is placed above other content in the tab order stacking context

- So screen readers announce the div as a native select element
- So the plain div can receive keyboard focus and fire the onKeyDown handler
- So React attaches the onKeyDown listener to the document instead of the element

Check answer

3. The dropdown is closed and highlightedIndex is 1. The user presses Enter once. What does handleKeyDown do?

- It does nothing, because Enter is only handled while the menu is open
- It opens the menu and also calls onChange with options[1]
- It opens the menu and returns immediately, without calling onChange
- It calls onChange with options[1] and leaves the menu closed

Check answer

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